15 Nov. 2018

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 55 and 56

ACME Advanced Combustion via Microgravity Experiments
E-FIELD Flames Electric-Field Effects on Laminar Diffusion Flames

E-FIELD Flames is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR).

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I. Introduction

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this experiment, an electric field with voltages as high as 10,000 volts is established between the burner and a flat, circular mesh electrode suspended above (i.e., downstream of) the burner. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a strong electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements are made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity allows for great simplifications in the analysis of ion-driven convection and flame interaction, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

An electric field can strongly influence flames because of its effect on the ions produced as a result of the combustion reactions. The induced ion wind can modify the flame shape, alter the soot and stability limits, and direct the heat transfer. The purpose of the E-FIELD Flames

experiment is to gain an improved understanding of flame ion production and learn how the electric field can be used to manipulate the ions and control non-premixed flames for practical benefit. For example, electric fields can enable stable burner operation at fuel-lean conditions with reduced pollutant emission, i.e., where the flame is not reliably sustained without the electric field. As another example, the role of electric fields on soot formation can be dramatic. It appears that the ion driven wind alters the local convection and strongly influences the residence time of the soot precursors within the hot flame. The experiment contributes to our critical understanding of combustion processes in the presence of electric fields.

More specifically, E-FIELD Flames has the following experimental objectives:

A. Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.
B. Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
C. Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
D. Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

The effect of the electric field on the flame is studied through tests conducted with a range of flame and electric field conditions. The fuel, fuel jet velocity, air flow, electric-field voltage and polarity are the primary parameters of the E-FIELD Flames experiment. Tests are conducted with methane and ethylene, including nitrogen dilutions of each, which provide a broad range of soot-producing conditions, i.e., from soot-free to highly sooting. They are conducted with both a simple gas-jet burner and the coflow burner that was also used for the Coflow Laminar Diffusion Flame (CLD Flame) experiment. Tests with the coflow burner is conducted with and without a coflow of air, where the results of the latter are compared with those from the gas-jet burner to assess the effect of air entrainment on the flame. The coflow burner also provides a different electrode configuration (close to a plane-to-plane) than the jet burner (close to a point-to-plane). In a typical test, the mesh electrode is energized at increasing potentials (relative to the burner), where the flame's response (e.g., ion current, luminosity, shape) to the resulting field strength is measured. The field strength is changed in repeated steps or is instead smoothly ramped.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Microgravity flame flicker is then the result of electric body forces alone. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low

velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application. Chemi-ion species, electronically excited chemiluminescent species are are very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry.

II. Methods/Research Operations

A. Discussion of Method/Protocol

E-FIELD Flames and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, the mesh voltage can be adjusted to approach a flame's stability or soot limit. This was not done during Increments 55-56, where the console commanding during a test was typically limited to halting the test when it was observed in the analog ops video that the flame had extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 142 flames were ignited for the E-FIELD Flames experiment during Increments 55 and 56. The first round of testing occurred from March to May 2018, which was followed by a second round, which began in September 2018 and was still underway at the beginning of Increment 57. The first round of tests resulted in 120 ignited flames

conducted with the same coflow burner used by the Coflow Laminar Diffusion Flame (CLD Flame) experiment, where CLD Flames's second round occurred between the two rounds of the E-FIELD Flames expeirment. The 'interruption' in the E-FIELD Flames testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

E-FIELD Flames does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH_4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C_2H_4), is highly sooting because of its carbon double bond. For both fuels, tests were conducted with pure fuel and fuel mixed with nitrogen over a broad range of dilutions. For limited cases, the fuel and nitrogen was premixed on the ground prior to launch. ACME bottles were launched with methane concentrations of 30%, 40%, 45%, and 100% with a balance of nitrogen (on a volume basis). Similarly, bottles were launched with ethylene concentrations of 30%, 50%, and 100%, again with a balance of nitrogen (on a volume basis). But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations. In all cases, the coflow was 21% oxygen with a balance of nitrogen (on a volume basis).

There are no samples (e.g., of soot) collected for E-FIELD Flames, where all of the ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the E-FIELD Flames ignited for E-FIELD Flames during Increments 55 and 56, including a limited overview of the test conditions, is found on the next page in Table 1.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55 and 56							
Test Day[1]	**Year**	**Month**	**Date**	**Mesh Polarity[2]**	**Fuel bottle**	**Fuel conc.[3]**	**Flames ignited[4]**
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Start of the second round of tests, with the gas-jet burner, in Sept.-Oct. (Increment 56)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

The primary hard drive for the CIR's IPSU-G Image Processing and Storage Unit - for Gigabit Ethernet (GigE) cameras – failed to boot on April 27, 2018 (18117). For ACME experiments, the IPSU-G is used for two key cameras, the ACME color digital 'data' camera (i.e., DataCam) and the Cool Flames Investigation (CFI) intensified camera, which is used for CLD Flame with a narrow bandpass filter to image CH* chemiluminesence. Use of the IPSU-G was recovered by astronaut Scott Tingle on May 7 (18127) by the removal of that primary drive. The removed drive was reinstalled on Aug. 14 (18226) by astronaut Ricky Arnold during the second round of CLD Flame testing. Although the hard drive was installed in the secondary position, the IPSU-G surprisingly again became unusable. Therefore, it was removed on Aug. 22 (18234) by astronaut Alex Gerst, returning it to use, albeit without the intended storage capacity.

The CIR's IPSU-1 Image Processing and Storage Unit, which is used with the High Bit-Depth Multispectral (HiBMs) camera and the collimated illumination package, was only intermittently operable in mid-Sept. 2018 and was not operable at the start of E-FIELD Flames' second round. Those capabilities were not seen as critical for E-FIELD Flames, where the illumination is significantly blocked by the mounting bracket for the mesh electrode. Therefore, the tests on Sept. 26 (18269) and Oct. 1 (18274) were conducted without HiBMs imaging or the CIR illumination. It was later determined to be an issue with a cable connection, which was corrected by astronaut Alex Gerst on Oct. 22 (18295) when he carried out other ACME tasks.

2. Any other difficulties in obtaining scientific data

E-FIELD Flames tests sometimes automatically abort (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time in the failed test attempt. The aborts normally occur because of communication issues with IPSU-1. These are infrequent, but there are some days when tests abort repeatedly.

Tests are of course also unsuccessful when the flame fails to ignite. The first test of the day has sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition is sometimes unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and synthetic air (i.e., 21% oxygen with a balance of nitrogen, on a volume basis) have unfortunately not always performed adequately. The fuel and especially nitrogen flows sometimes oscillate. Furthermore, the MFC used for the 'air' coflow behaves as if it is sticking in a few places, where the flow rate won't linearly ramp when commanded (as required for the CLD Flame experiment). It tends to stick in some places and then abruptly overshoot where that has sometimes been seen to extinguish the flame. As a result, mass flow controllers are tested - without flames - after they are first installed. Adjustments are often then made to (1) the settings that control when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This activity has involved significant testing, separate from flame tests, and has taken place during CLD Flame operations on 17333, 17334, 17347, 17354, 18008, 18022, 18038, 18046, 18143, and 18241. In this regard, there are an assortment of MFCs with different flow ranges for each of the three gases. While the MFC tuning

has been beneficial in some cases, it has not eliminated all gas flow problems. It was seen that nitrogen oscillations tend to damp with time, so a nitrogen pre-flow was implemented so that the oscillations would typically end before the flame measurements. That approach has been partially successful and was introduced during E-FIELD Flames' first round. Later, a procedure of re-zeroing the MFCs prior to each test was also implemented during E-FIELD Flames' second round. A replacement MFC for the 'air' coflow is being prepared for spaceflight, where a limited third round of CLD Flame testing is being planned to accomplish its extinction limit objective.

The final day of testing during E-FIELD Flame's first round was conducted with highly sooty ethylene flames and a positively charged mesh, which tends to push the flame down toward the burner. A significant amount of soot deposited on the burner, both on the inner tube from which the fuel issues as well as the honeycomb in the burner annulus from which the air flows. The extent of the soot wasn't appreciated until flame testing for CLD Flame's second round began on May 25 (18145). Additional evaluation tests were performed on May 31 (18151), and the decision was made to have the crew clean the soot from the burner because it was clear that the gas flow from the burner was affected by the soot buildup. A cleaning procedure was developed, approved, and carried out by astronaut Ricky Arnold on June 19 (18170). The cleaned burner was imaged with the CIR cameras on June 21 (18172), but the evaluation testing was deferred because of a cable mating issue. The cable mating was corrected by Arnold on June 25 (18176) and the evaluation testing of the cleaned burner was conducted on the following day, June 26 (18177). Regular flame testing in CLD Flame's second round finally commenced on July 5 (18186), six weeks after the soot accumulation was identified as an issue. However, useful scientific results were obtained on the June 26 evaluation day, so the burner cleaning effectively led to a 1-month delay.

C. Completeness/Quality of Data

The number of flames ignited significantly exceeds the planned tests which were (1) specified as required in the ACME Integrated Science Requirements Document (ISRD), using the gas-jet burner, and (2) later expanded to include tests with the coflow burner to partially make up for science loss resulting from the slow response time of the high-voltage power supply. Following the revision, a total of 74 E-FIELD Flames tests were identified as required. In contrast, 120 flames were ignited in the experiment's first round, i.e., first half. In that round, roughly 200% more flames were ignited than had been planned.

It should be understood that a complete set of image data is not available for all tests. As discussed in section III.B, the IPSU-1 was not working at the beginning of the second round of testing, preventing use of both the HiBMs camera and the CIR's collimated illumination package. Those are not critical for E-FIELD Flames, so tests continued without their use.

Furthermore, recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and E-FIELD Flames is considered to be highly successful.

D. Status of Data Analysis

Data from the E-FIELD Flames experiments includes video recordings from the operations camera for general flame behavior, the ACME data camera for high resolution images at various exposures and frame rates as dictated by the flame luminosity, intensified camera images and non-imaging data at 100 Hz. The non-imaging data is extracted using Matlab scripts (modified from those developed by CLD Flame researchers) to show trends of ion current and flame luminosity as a function of the applied field, and the flame parameters (fuel, flowrate, burner type). Image data is processed using the OMA software (also developed and used by CLD Flame researchers).

The overall data volume is large with a storage limit of 127,000 MB per test day, the data has been downloaded to the hard drive with a backup system at UC Irvine. At this stage, with operations ongoing, the analysis is comparative and qualitative, just to show the major differences in behavior in microgravity and normal-gravity (1-g) flames.

E. Preliminary Research Findings

Results from the early E-FIELD Flame experiments have demonstrated clear differences between normal-gravity and microgravity flame behaviors. A few examples include the distinct maximum of ion current for co-flow burner flames in microgravity, where the 1-g flames more smoothly reach a saturation condition. The results also show that the electric body force can produce flame oscillations similar to those driven by buoyancy in normal gravity, and microgravity flame shapes with an electric field can mimic normal-

gravity flames without electric fields. Burner type plays a role in the flame/field interaction, and in particular, reversing the field polarity for the jet flame does not produce an inverted result over the entire voltage range tested. Results also demonstrate the interactions between electric fields and soot are more complex in microgravity. At the same time, the preliminary findings suggest that some of the key expectations generally hold (e.g., ion current scales approximately with flow rate for a fixed fuel and burner).

100% methane for positive field strength with and without coflowing air

This section only includes representative findings as the ISS experiments covered a wide range of conditions and outcomes. Color camera images, intensified camera images, total flame luminosity, and ion current, along with all test parameters and flow monitoring, were recorded in all cases. Figure 1 gives example flame images from the first tests conducted with methane as fuel. Two fuel flow rates are shown, with the voltage indicated as nominal field strength (kV/cm). Since the physical spacing from the burner to the mesh is 37 mm, 2 kV/cm represents 7000 V across the flame.

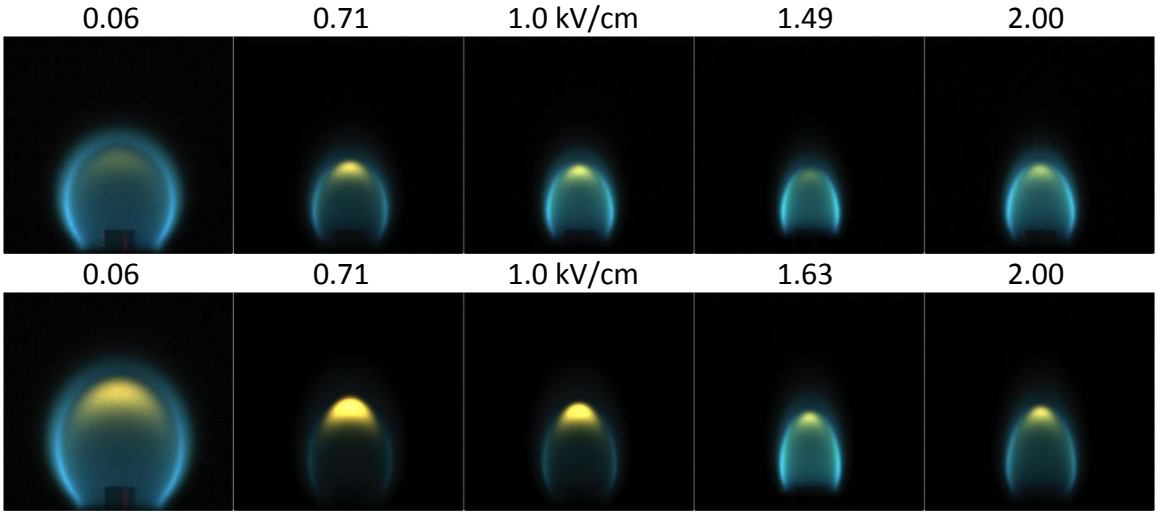


Figure 1. Typical flame appearance as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

The key measurement for the E-FIELD Flames experiments is the ion current as a function of applied voltage, i.e., the voltage-current or V-I curve. This curve provides information on the flames ability to produce chemi-ions based on its saturation ion current, and it provides a measure of combustion performance in that ion production is a function of carbon influx into the flame and the vigorousness of the combustion process. More compact, higher temperature flames produce more chemi-ions than do larger more diffuse reaction zones of lower overall temperature for the same fuel flow. The first effect (i.e., the relationship between carbon influx and peak ion current) can be seen in Figure 2, where the flow of 15 cm/s methane produces slightly above 2 microamps of ion current at the peak (1.5 kV/cm) and the flow of 19 cm/s methane has a peak of 2.75 microamps. While the ratio is not exact, the peak ion current is nearly proportional to the flow

velocity. Another interesting feature of these curves is the peak value. Traditional electrical aspects of flames studies look for a saturation ion current plateau, where the ion current remains fixed despite an increase in applied electric field. For these microgravity flames, there is a pronounced peak in the ion current at saturation and then a gradual decrease to a global saturation plateau. This is a strikingly different behavior than is seen in 1-g, and the behavior is even more pronounced for the case where there is zero coflow.

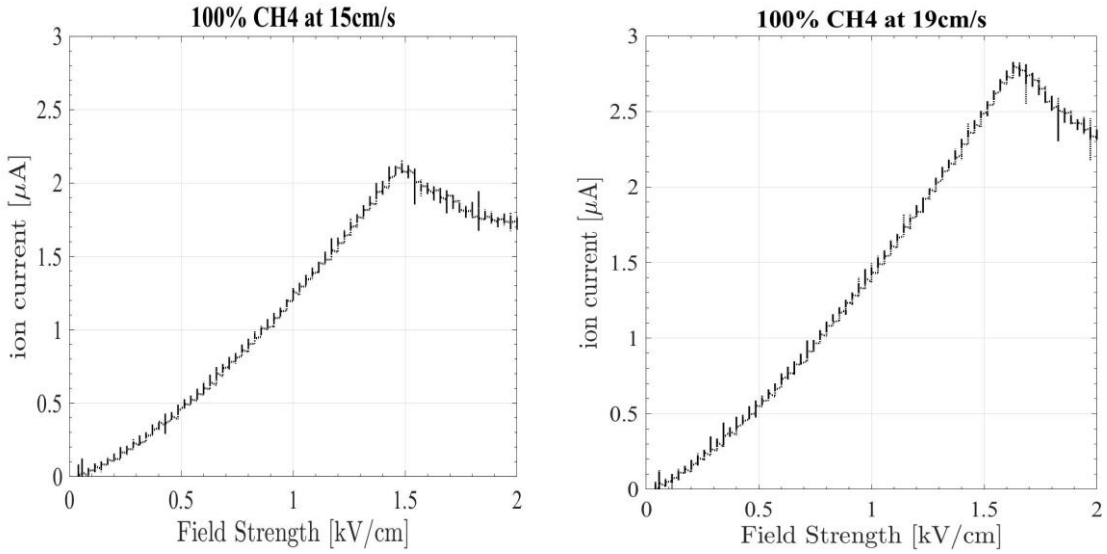


Figure 2. Ion current as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

As can be seen in Figure 3, which is the same V-I curve result in microgravity for the case where there is no coflowing air, and Figure 4 shows the flame images. These figures again show a reasonably linear scaling of peak ion current with fuel flow velocity, but they demonstrate a very pronounced peak ion current with a rapid drop to a global saturation ion current plateau. Since these are diffusion flames, essentially all of the fuel is being burned independent of the flame's shape and the applied electric field, but it seems clear that the electric field is affecting the chemi-ion production capability of the flame. As is apparent from the flame images, the flame is most compact at the voltage consistent with the peak ion current. With increasing voltage, the flame becomes less compact and the ion current decreases. We presume that the flame remains at saturation throughout this period (that is, the flame is producing as many ions as it can, and those ions are being removed from the reaction zone as soon as they are created) so the decrease in ion current results from a change in combustion intensity. We can explore this hypothesis further using the PMT data collected in the experiments.

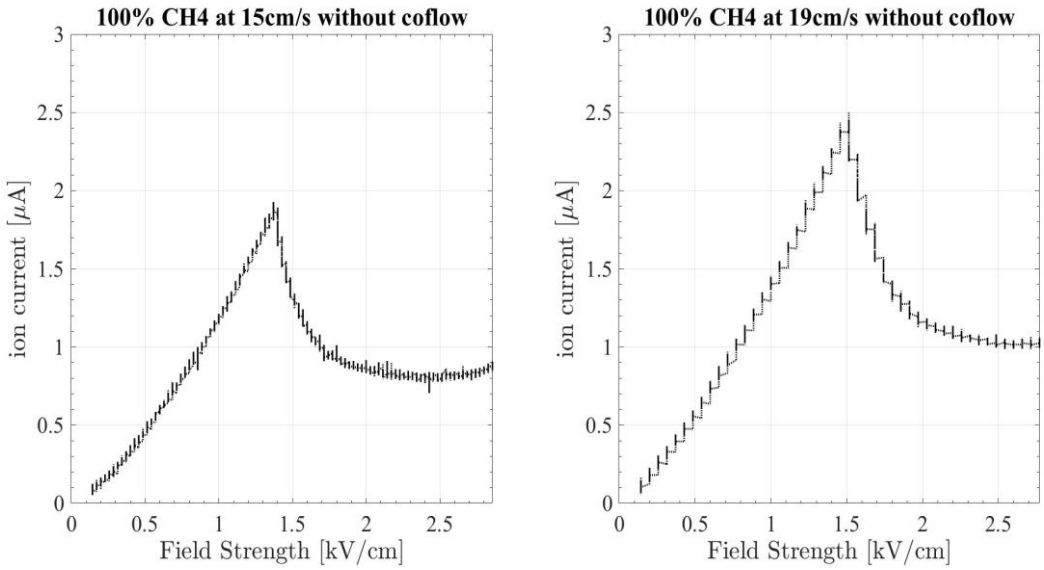


Figure 3. Ion current as a function of field strength and fuel velocity with no coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

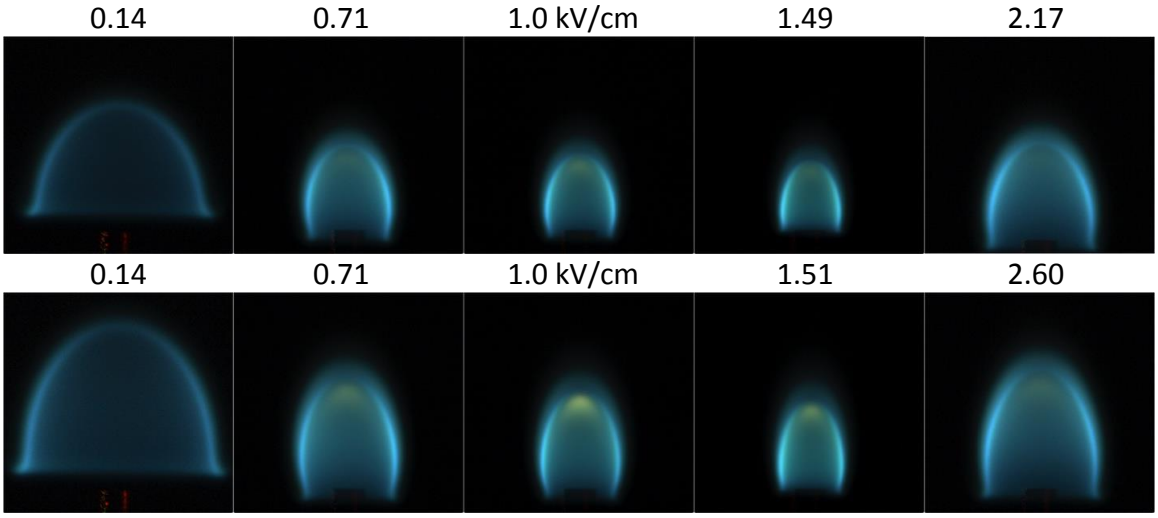


Figure 4. Flame appearance as a function of field strength and fuel velocity with no coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

In all of the curves, the jagged features are spikes following a voltage step change that are mostly associated with the transient capacitance of the system when rapid steps are requested. These overshoots notwithstanding, the V-I curves follow smooth and reproducible trajectories.

For comparison, the equivalent V-I curves for 1-g flames with the same burner configuration with no coflow are shown in Figure 5. The results are too recent to allow a detailed analysis and discussion of the differences, but some significant observations from these results and the comparison follow.

(1) The ion current reaches a peak at approximately the same field strength for both velocities in 1-g while in microgravity the ion current peak for a lower fuel velocity occurs at a lower field strength. The field strength at the peak current seems to scale linearly with the fuel velocity.

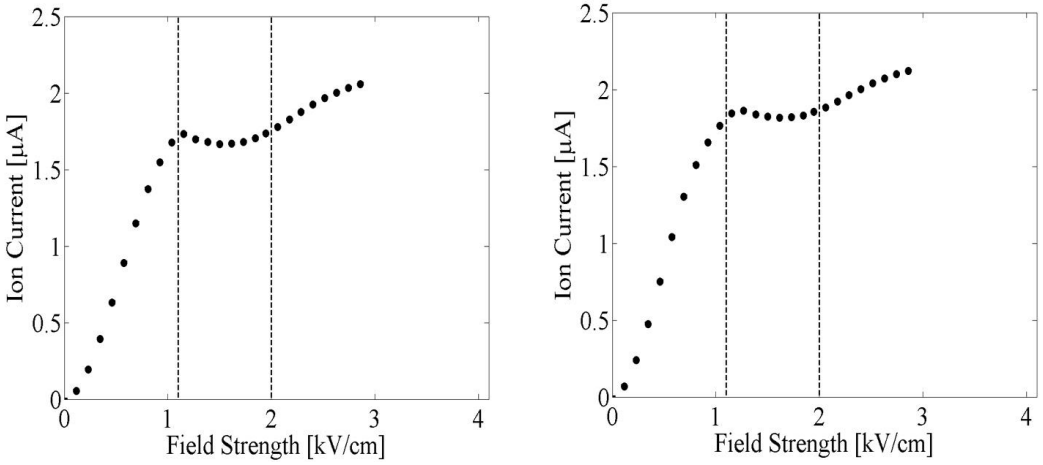


Figure 5. Ion current as a function of field strength and fuel velocity with no coflow in normal gravity. (a) 16 cm/s methane. (b) 19 cm/s methane.

(2) In microgravity, the ion current does not smoothly increase to saturation nor does it show a gentle peak with a mild decrease as is seen in the normal gravity results. Instead, in microgravity there is a distinct peak in the ion current with a severe decrease throughout the saturation condition, with a final saturation plateau observed at voltages substantially above the voltage at peak ion current. That is, the microgravity final saturation is not the maximum ion current observed. This difference is more dramatic when there is no coflow, which suggests that entrainment plays an important role whether caused by buoyancy or a forced coflow.

(3) The peak ion current is higher for microgravity than comparison 1-g flames. The reason is not obvious at this point but may result from a more focused upward flow from the electric body force and ion-driven wind as compared to the combined effects of the ion wind and buoyancy.

(4) The peak ion current is approximately proportional to fuel flow velocity, which, as described in prior studies, reflects the relationship between ion production and the carbon flux into the system. The peak ion current also increases with the coflow which suggests that healthier oxidizer transport helps increase the ion production.

30% ethylene with coflowing air at 3.1cm/s

The results from ethylene flames with coflow are shown in Figures 6 and 7, positive field and negative field respectively. At 15 cm/s, the ion current reaches a peak around 2.7 microamps which is similar to the 19 cm/s 100% methane coflow air case but with only 0.4 of the volumetric carbon content. The results for 15 cm/s also presents a rapid parabolic behavior after reaching the saturation peak. In the negative field case, the peak ion current decreases to 0.56 of the positive field case at the same flow rate 19 cm/s (or 0.4 for 15 cm/s). Another aspect is that the ion current peaks sooner for the higher flow rate/carbon content in the negative field configuration as compared to the positive field experiments. These phenomena, particularly for the positive field strength have not yet been measured in the laboratory at 1-g. The difference in behavior appears to be related to the removal of the buoyancy force so that the momentum flow coupled with the ion driven flow tends to resettle under the electric field body force. Detailed analysis is underway.

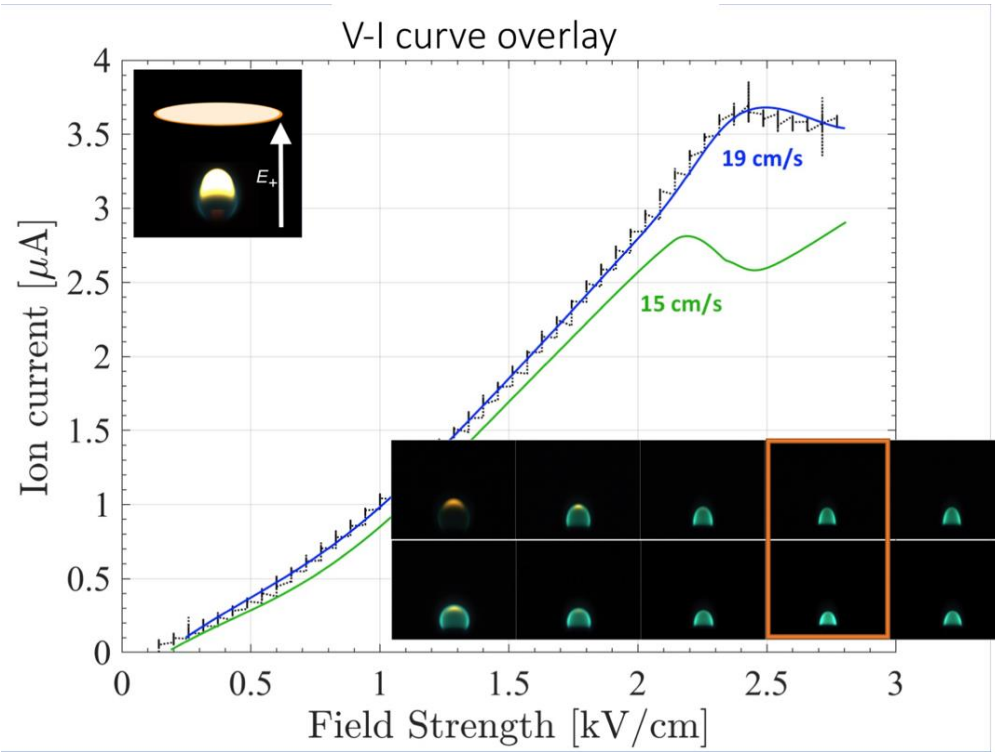


Figure 6. Ion current to electric field strength per unit distance for positive field of 100% ethylene with coflow air. The ion current peaks around 3.5 microamps at 19 cm/s and 2.7 microamps for 15 cm/s.

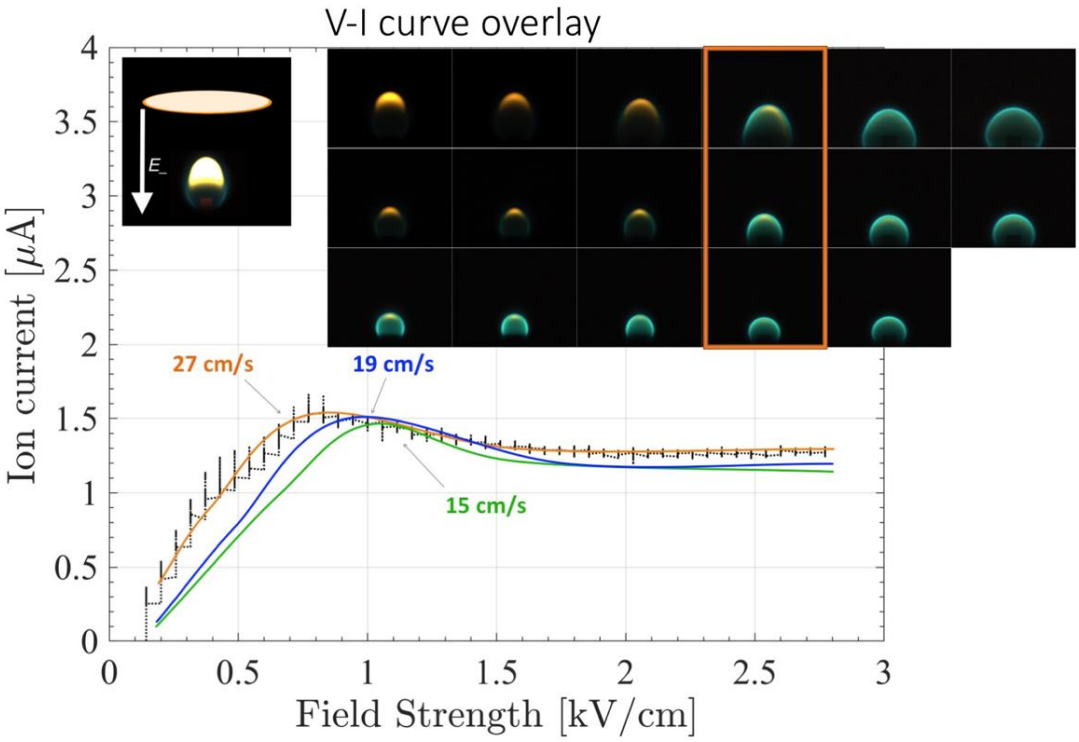


Figure 7. Ion current to electric field strength per unit distance for negative field of 100% ethylene with coflow air. The peak ion current reduces to around 1.5 microamps for all the fuel flows: 27, 19, and 15 cm/s.

Early E-FIELD Flame results from the ISS testing were presented in following oral and poster presentations:

1. Derek Dunn-Rankin and Yu-Chien (Alice) Chien, "Electrical Field Effects on Laminar Diffusion Flames," 11th Multi-Agency Coordinating Committee for Combustion Research (MACCCR) Annual Fuel and Combustion Research Review, Livermore, CA, 9-11 April 2018. (oral presentation)
2. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Electric Field Effects on Flames in Microgravity on the International Space Station," Central States Section Combustion Institute Spring Technical Meeting, St. Paul, MN, 20-22 April 2018. (conference paper and oral presentation)
3. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Small Diffusion Flames under the influences of Electric-Field: Microgravity and Gravity," 37th International Symposium on Combustion, Dublin, Ireland, 29 July to 3 Aug. 2018. (Work-in-Progress poster)
4. Yu-Chien (Alice) Chien, Jesse Tinajero, Dennis Stocker, Uday Hegde, Derek Dunn-Rankin, "Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames," 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) Meeting, Rockville, MD, 31 Oct. to 3 Nov. 2018. (oral presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Meanwhile, E-FIELD Flames videos can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/e-field-flames/

Videos related to E-FIELD Flames and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.